Advanced Embedded Systems Lab

Lab 01: Introduction to Arduino Wifi

Hochschule Hamm-Lippstadt
Summer Term 2025
05.05.2025

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Agenda

1. **Organization**
2. Arduino
3. Arduino Uno WiFi Rev 2
4. First Steps of Arduino Programming
5. Lab 01

1. Organization

- Will be uploaded soon.

1. Organization

- Labs and Projects submission:
  - ~~Team selection~~
  - If you still don’t have a team please send an email!
  - Each team organizes the labs and project in <u>one</u> Github repository
    - Set up a Github repository
    - Send me an invitation (Ali.alhalabi@hshl.de)
    - Upload your labs and project documentation to your GitHub repository

1. Organization

Hardware

- Each team will receive its own hardware
- Each team is responsible for its own hardware
- Each team should return the hardware (without defects) after the final presentation.

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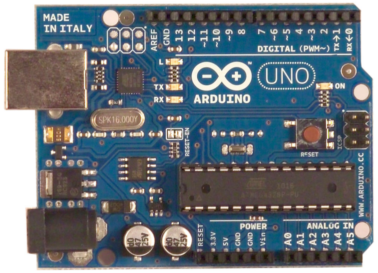
2. Introduction to Arduino

- Easy-to-use, well-supported and inexpensive development board
- Simple and accessible user experience
- Do not require extra hardware to program and use the microcontroller
- Wide variety of C/C++ functions that work with all the “Arduino family”
- Open-source
- Huge amount of online documentation (projects, samples, etc)
- Large amount of “shields” that you can add on to do more things
- Great for prototyping ideas

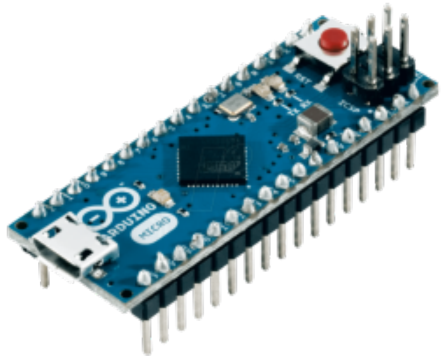
2. Introduction to Arduino

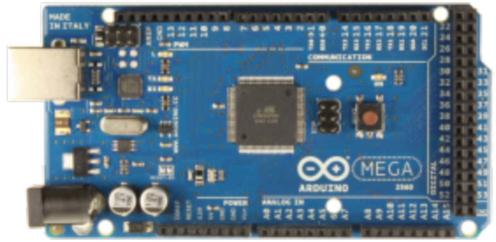
Arduino shields



Arduino Uno

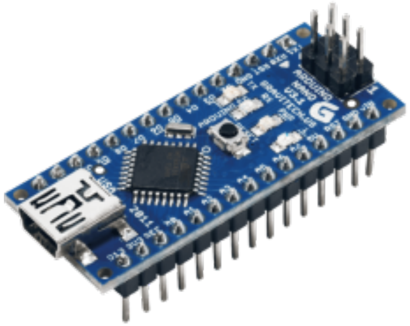

Arduino Uno Wifi Rev2

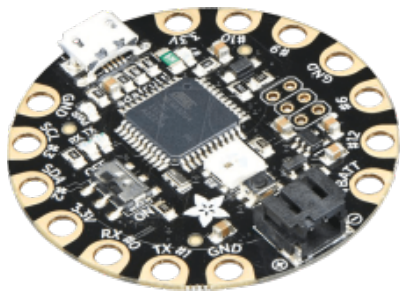

Arduino Micro

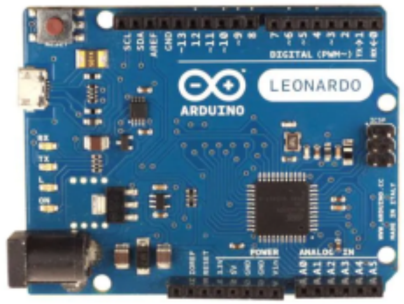


Arduino Mega


Arduino Ethernet


Arduino Nano


Arduino Lilypad

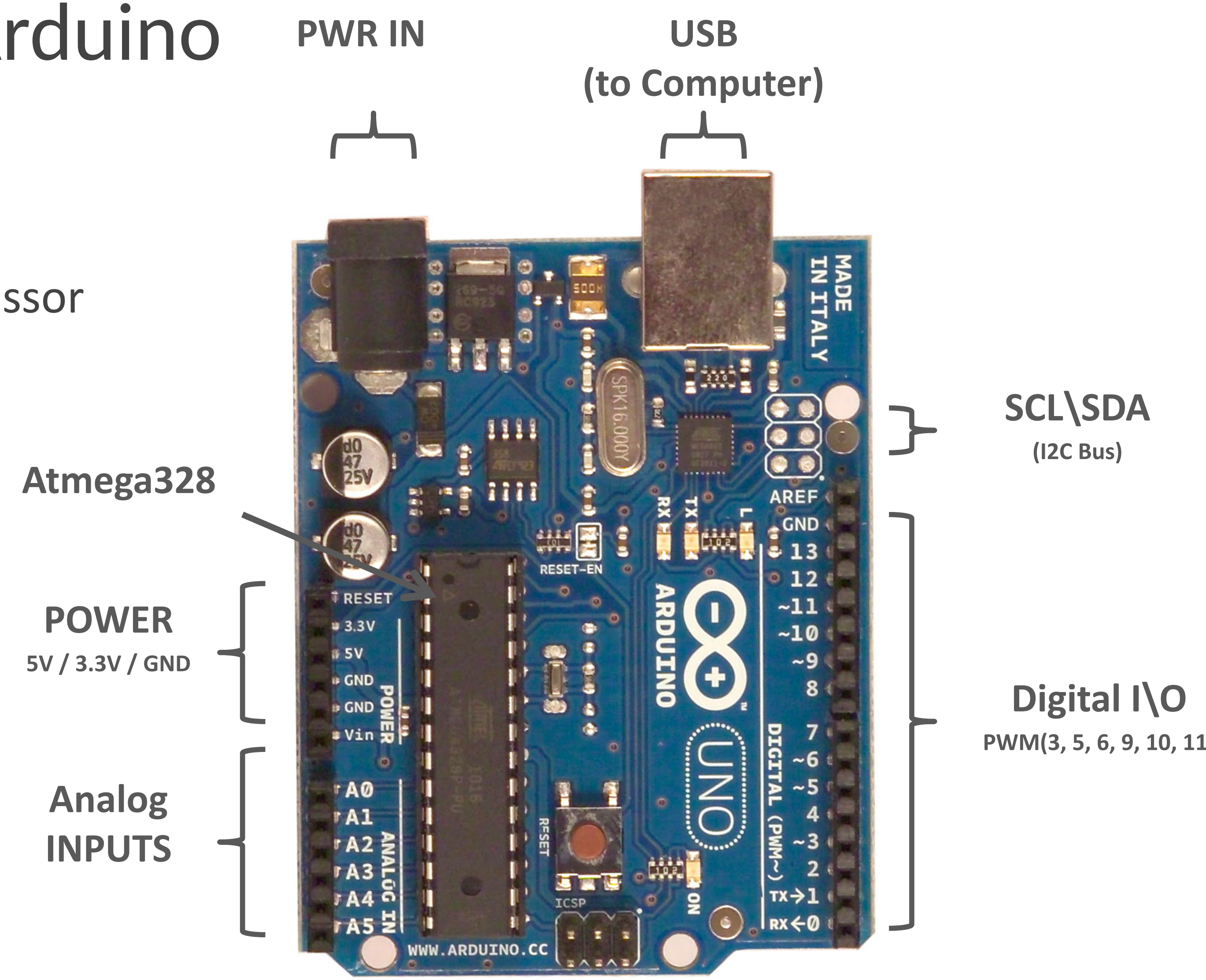


Arduino
Leonardo

2. Introduction to Arduino

Arduino Uno

- 8 bit Microchip microprocessor (ATmega328)
- 14 digital I/Os
- 6 PWMs
- 6 analog inputs
- Serial interface



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3. Arduino Uno Wifi (R2)

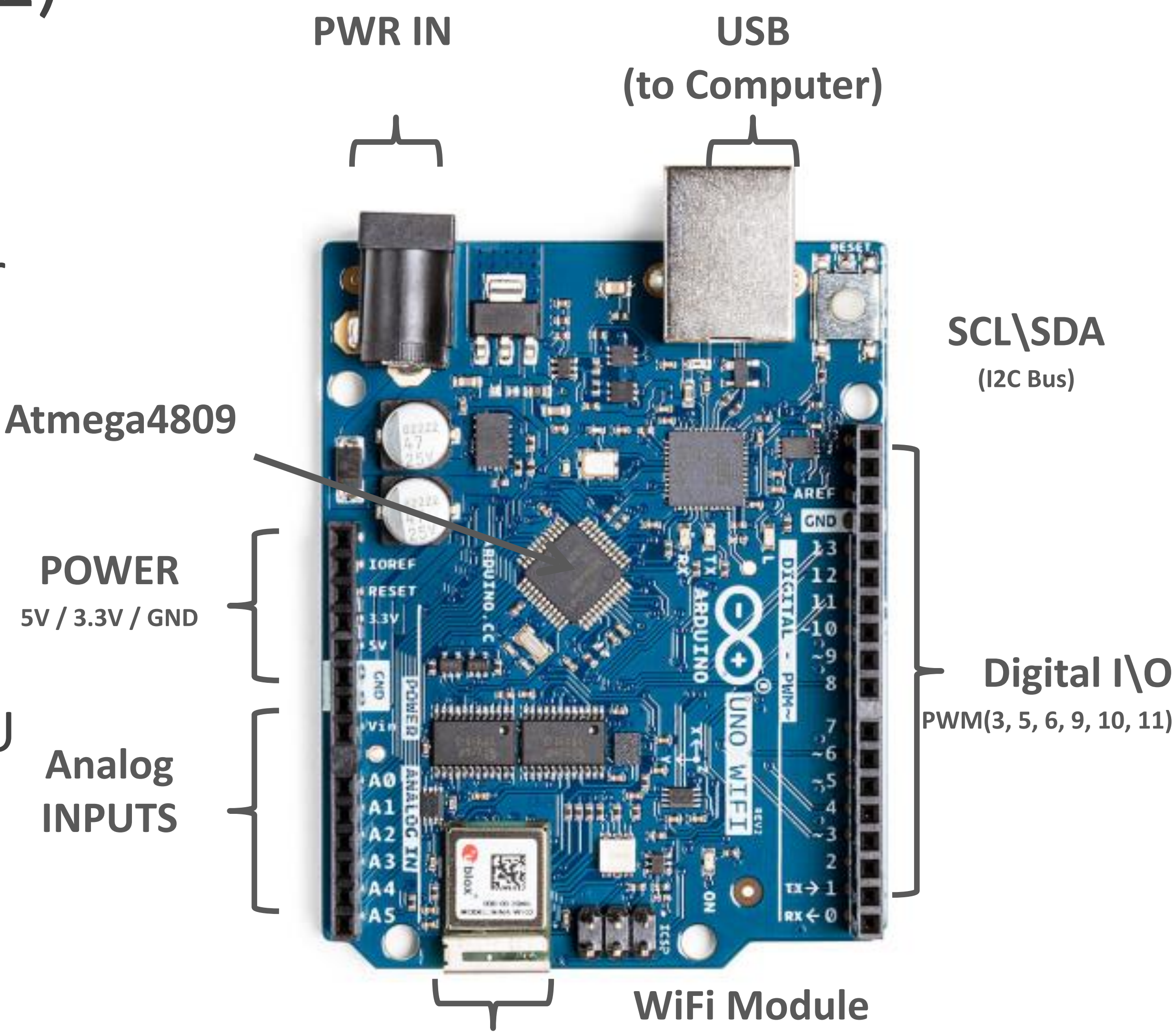
Arduino Uno Wifi (R2)

- The Arduino Uno WiFi Rev2 is an **Arduino Uno** board with an **integrated WiFi module**
- The board is based on the Microchip **MEGA4809** with an **ESP32 u-blox NINA-W13 WiFi Module** integrated.
- The NINA-W13 Module is a self contained SoC with integrated TCP/IP protocol stack that can give access to your WiFi network (or the device can act as an access point).
- The Arduino Uno WiFi Rev2 is programmed using the Arduino Software (IDE), our Integrated Development Environment common to all our boards and running both online and offline
- The Arduino UNO WiFi Rev.2 is the easiest point of entry to basic IoT with the standard form factor of the UNO family

3. Arduino Uno Wifi (R2)

Arduino Uno Wifi (R2)

- 8 bit Microchip microprocessor ATmega4809
- 14 digital I/Os
- 5 PWMs
- 6 analog inputs
- Inertial Measurement Unit IMU
- Wifi



3. Arduino Uno Wifi (R2)

Arduino Uno Wifi (R2) is not simply equal to Arduino Uno + Wifi

- Same form and size, but
- Different Microcontroller
- Some slight changes
- Sketches and Projects will not run automatically

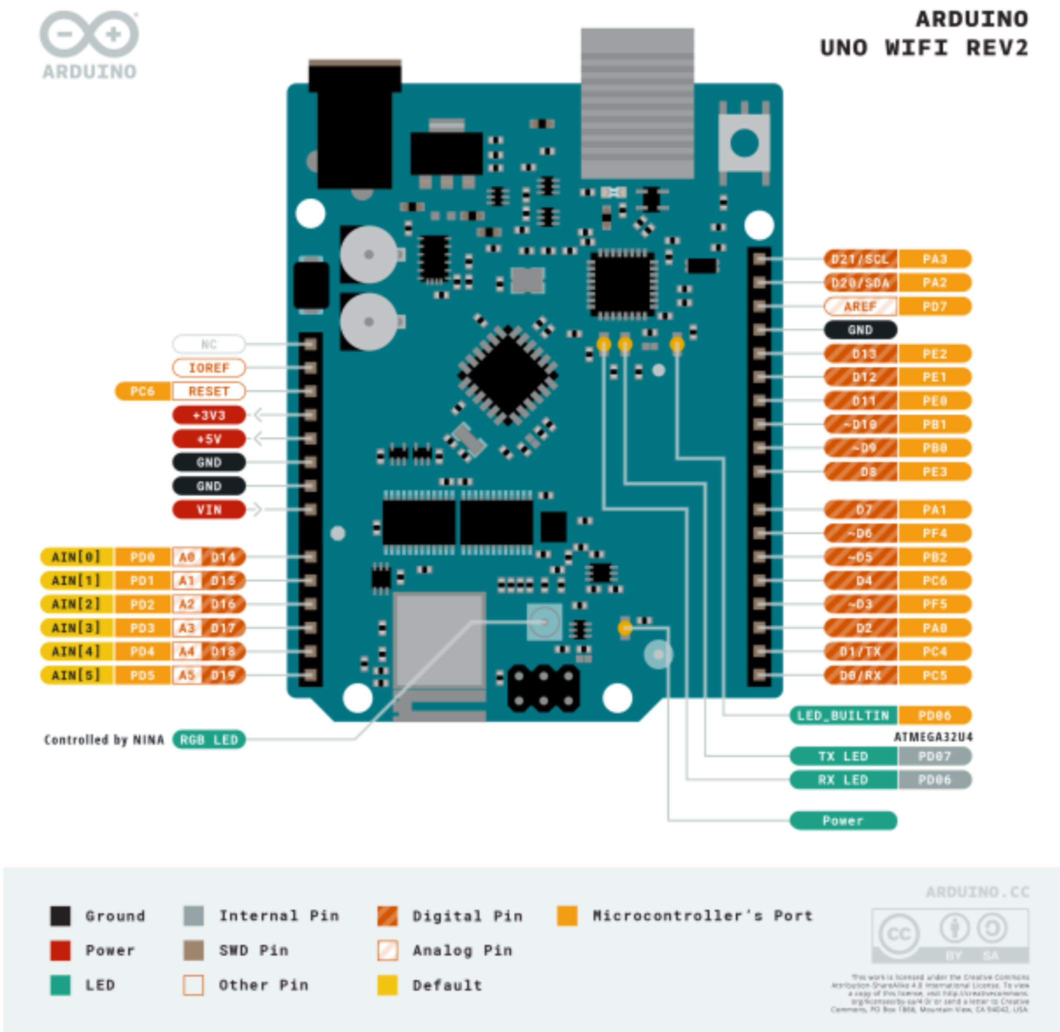
	Arduino UNO R3	Arduino UNO WIFI R2
Mikrocontroller	ATmega328P	ATmega4809
Clock frequency	16 MHz	16 MHz
Flash memory	32 KB	48 KB
SRAM	2 KB	6 KB
EEPROM	1 KB	256 KB

3. Arduino Uno Wifi (R2)

Arduino Uno Wifi (R2)

https://store.arduino.cc/products/arduino-uno-wifi-rev2#

Arduino Uno WiFi (R4) available!



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4. First Steps

Getting started with Arduino Uno Wifi Rev 2

- Download Arduino IDE (latest version)
  https://www.arduino.cc/en/software
- Arduino Uno WiFi R2 Configuration:
  - https://www.arduino.cc/en/Guide/ArduinoUnoWiFiRev2
  - Tools —> Board —> Board Manager —install—> Arduino megaAVR Boards
- Data sheet:
  - http://ww1.microchip.com/downloads/en/DeviceDoc/ATmega4808-4809-Data-Sheet-DS40002173A.pdf
  - https://content.arduino.cc/assets/Arduino_NINA-W10_DataSheet_%28UBX-17065507%29.pdf
- **First Task:**
  - **Try the blink example on your Arduino!**

5. Lab 01

1. Download the Wifi Library
   - Tools —> Manage libraries —install—> WiFiNINA
2. Understand the Wifi Library
   - https://www.arduino.cc/reference/en/libraries/wifinina/
3. Explore the features of WiFiNINA library and create a Wifi communication with your Arduino Uni WiFi Rev 2
   - https://docs.arduino.cc/tutorials/communication/wifi-nina-examples/
4. Control the on-board LED(GPIO 25) wireless using a web server.

Thank you for your attention!